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Mining during the next 25 years: issues and challenges

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This article examines possible future trends in the mining industry in the light of past developments. After a preliminary discussion, the article considers various aspects of demand, supply, corporate structures, geographic distribution of investment, and social and environmental factors influencing the minerals industry. Scepticism is expressed about the smooth, exponential growth envisaged by many commentators. Some risks are elaborated, and a digression about gold is included.

The article argues that, while the minerals industry has concentrated on reducing cash costs through technical innovation and productivity improvement, new approaches may be needed in future that are not directly linked to economies of scale. Contracting out is an interesting trend with major implications.

The view that the mineral industry's corporate structure is becoming increasingly concentrated is challenged. Developments in consumer countries as well as historical precedents strongly suggest a more diverse structure.

The spread of minerals investment from North America and Australasia, and the resurgence of foreign direct investment in minerals projects, are placed in context and discussed from the viewpoints of both companies and host countries.

Although the appropriate responses to many of these issues can only properly be made by governments at all levels, the industry has to play its part.

Fashion, though Folly's child, and guide of fools,
Rules e'en the wisest, and in learning rules.
George Crabbe, *The Library*, 1781

The author's intention in this paper is to be provocative, and to challenge aspects of the conventional wisdom about present and future trends in the global minerals industry. This means that questions may be asked, but answers not necessarily provided.

The future cannot be looked at in isolation, but should be seen as a series of frames in a feature film that has already been running for some time. What happens in the future is largely conditioned by the past, and cannot be completely divorced from it. The path into the future is often perceived as a continuous, if winding, road. It is suggested here that it might be more similar to a multi-centred maze with many dead-ends and unexpected turns. As we progress through it, we often observe features that we think we have seen before, but that does not necessarily guide us to the centre.

Many aphorisms observe that history repeats itself. In the minerals sector, there have undoubtedly been patterns of advance and retreat in all aspects of the industry, even within the post-war decades.

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These, sometimes repetitious, movements are often disregarded by the managements of mining companies. Thus, intent on avoiding the apparent mistakes of their immediate predecessors, they possibly fall into the same errors as earlier generations. The industry's corporate memory has been greatly truncated in recent decades by the universal slimming of head office functions, and by rigorous cost-cutting. These circumstances increase the likelihood of the industry repeating history, whether tragically or farcically.

The present article examines a variety of different strands that will be woven together to create the industry's future. Some are fairly distinct, but others closely intertwine and influence each other, which should become more apparent from the following discussion.

All too often, the mining industry is described in terms of two metals, gold and copper, rather than as the multi-faceted and multi-product industry that it really is. Certainly the gold companies have in the past decade or so been the storm troopers who have spear-headed the industry's geographical diversification, with copper acting as the main tank brigade, but concentration on these two products can unduly narrow the focus, and possibly even obscure some trends. Another undue simplification is to treat the global mining industry as the exclusive preserve of multi-national companies, and to overlook the significant contributions made by national companies.

Before looking forward, it is salutary to look back at the projections and prognostications made 20 or 30 years ago. Few have stood the test of time. Most have reflected the present, and immediate past, circumstances of the time at which they were made. That is especially true of forecasts of demand, costs, and prices. The widespread concerns about security of supply of the late 1970s have evaporated with the changed global political environment, and have given way to an undue complacency, at least in many of the mature industrial economies. Likewise, the quest for means of stabilising mineral prices and producers' incomes has apparently foundered. Will today's policy concerns about, for example, foreign investment, environmental protection, and the local community prove any more durable? In the late 1970s it was widely asserted that privately-owned mining companies were being squeezed out between the millstones of state ownership and dominance by the oil companies. It was also claimed that investment would be increasingly concentrated in an ever-narrowing group of countries, with adverse effects on the future availability of minerals. Yet, the world looks rather different today.

By contrast, the report on Mining and the Environment commissioned by a group of mainly British mining companies in 1972, known as the Report of the Zuckerman Commission, is as relevant today as when it was published in 1973. Some trends may persist, but many other apparently inexorable developments carry the seeds of their own demise and reversal. In short, we should be humble when looking ahead, and ensure that all our policies, whether corporate or governmental, are appropriately flexible.

So much for the preliminaries. Let us now turn to some of the issues and challenges.

Future issues affecting demand for mineral products

The future of the global minerals industry ultimately hinges on the behaviour of demand for its products. Without that demand, there would be no mining. The first point of reference here is that any precise forecast will be wrong, whether it is the acute pessimism of the early to mid 1980s or the relative optimism of the mid 1990s. As already noted, most forecasts are grounded in the recent past. Most of the minerals industry completely ignores any explicit forecasts of demand and operates on a variant of Say's Law, that supply creates its own demand. Many companies are technologically driven and obsessed with their relative costs. They sometimes forget that others are following similar policies which may work temporarily, but which do nothing to ensure that there is always enough demand to absorb ever-expanding supplies.

Will demand continue to grow?

Will total demand for mineral products continue to expand exponentially? That partly depends on the future progress of economic activity. Most medium to long term forecasts, such as those of the World Bank or the IMF, invariably predict a smooth economic expansion well into the next century. This view contrasts sharply with the erratic volatility of the past. It is admittedly difficult, if not impossible, to predict the timing or extent of any future business cycles. But it would be foolish to deny their inevitability as the forces which generate them are deeply embedded in human behaviour. Future global expansion will not just spring from continued rapid growth in Asia and the expected revival of the transitional economies and Latin America. Those regions are perhaps even more prone to cyclical swings than mature industrial nations. Most current forecasts assume a 'surprise-free' world without any major political upsets or other discontinuities. Surprises will, however, inevitably happen, and they will be more likely to affect the faster growing regions.

Demand in relation to economic growth

Even if economic growth does remain strong, will this translate into steadily rising demand for mineral products? The Club of Rome's gloomy predictions have by now largely been dismissed. But might they not contain some important truths? Take copper as an example. Many commentators expect an annual average growth rate of demand of around 3% per annum. That implies that the world's usage will rise from roughly 12 million tonnes today to nearly 14 million tonnes by 2000, over 18 million tonnes by 2010, and 25 million tonnes by 2020. What does this, in turn, imply for capital investment in mines and processing plants, and for relative prices? A more immediate question: Is such prolonged growth probable? Copper is here taken as an example for all mineral products.

Impact of technological change

Although technological change and environmental policies, including recycling, will remain important factors influencing demand, it is always easy to exaggerate the likely impact of technological change and substitution on the existing uses of any product. This is partly because defensive retaliation is overlooked. Also there is a natural tendency to assume that innovations will be introduced and diffused more quickly than what usually happens in practice. Consequently, it can be observed that metals have retained and expanded their markets in the face of plastics and composites.

It is also too frequently assumed that all countries will follow similar patterns of development, and that per capita levels of consumption of individual materials will rise to those currently prevailing in the United States. Such assumptions ignore the wide-ranging differences among countries in their natural resources endowment, comparative advantages, and economic structures. Also they overlook the tendency for each new investment to embody the latest available technology, and for economies to leapfrog over whole stages of development. A pertinent example, for copper, is the rapid growth of mobile telephones in industrialising countries in preference to cable-based systems. In short, the future economic expansion of developing countries is unlikely to be as materials-intensive as that of the more mature economies. Moreover, the materials used will often be those lying at the technological frontiers of demand in the latter countries. These will often be plastics.

One of the major influences on demand for minerals, and for metals in particular, throughout the world will be the development of personal transport systems. How will the automobile industry develop over the next 25 years? Will the use of cars be greatly circumscribed in many nations? Will the internal combustion engine be replaced by a different means of propulsion? There is a variety of possible outcomes with differing implications.

The influence of oil prices

Economic forecasts are agnostic about future trends in prices of energy, and especially of crude oil. Although any sudden and unexpected shift in oil prices is unlikely to have the same dramatic effects on global activity as those of the 1970s, it would affect the distribution of activity. Many of the faster growing economies are heavily dependent on imported energy. Moving specifically to the minerals industry, the growth of demand for its products has been inversely related to trends in real oil prices. Much of the falling intensity of use that occasioned so much concern in the early 1980s can be traced back to the effects of rising oil prices. Conversely, the weaker tone of oil prices since the mid 1980s has facilitated faster growth. Is it safe to assume that oil prices will remain as benign in the coming two decades as in the last?

Environmental concerns

Demand for mineral products is being increasingly influenced by environmental considerations. These are expressed in many instances through changes in the relative shares of primary production and recycled material, whether direct, as in steel, lead or aluminium, or indirect, as with the zinc and copper in brass, the nickel in stainless steel, or the soda ash in glass cullet. Irrespective of the relative operating costs of the various sources, there is strong political and social pressure to increase the share of recycled products. Recycling will often lower the observed rate of growth of demand for the primary materials.

Effect of restrictions

Restrictions on the use of particular materials in specified uses have even more direct effects on demand. These can wipe out complete markets almost overnight, as has happened, for example, with cadmium in pigments, or mercury and asbestos generally. For example, the global output of asbestos has roughly halved over the past 15 years, whereas forecasts in the 1970s were for continued expansion. Markets for many metals are currently threatened for environmental reasons, particularly in northern Europe and the United States. The fact that the scientific basis for these threats is often shaky at best does not minimise their strength or relevance, as public policy is more often driven by emotion than by science. Also, producers of competing materials naturally exploit fears about metals to their own advantage. Environmental legislation or other restrictions against materials applied in the mature industrial countries quickly communicate into similar measures elsewhere. Unless the producers of threatened materials respond vigorously, future rates of demand could be much lower than expected. Yet, often the mechanisms for such a response do not exist, and many producers appear unconcerned about markets, as already noted.

Demand for gold

Before we leave demand, a few remarks about gold are warranted, because of the role that gold production plays in the global development of the minerals industry. It is almost an article of faith that Asian demand for gold will expand rapidly, and provide the foundation for rising mine supply.

Is this necessarily true? Why should gold not experience the same technological and economic leapfrogging by the developing countries that producers of all other materials accept as inevitable? Cultural factors do change with incomes and technology. The development of more sophisticated banking systems in the Asian economies, coupled with improved political and personal security and rising living standards, could reduce Asian demand for carat jewellery used as a store of value by more than the offtake of jewellery for adornment increases. If that did happen, and it were accompanied by a continued drawdown of the hoards of central banks and private individuals, gold would behave more and more like other metal commodities. This is not a forecast, but the depiction of one plausible outcome that emphasises the mining industry's need for flexibility in the face of an uncertain future.

Centre of gravity of demand

The one certainty is that the geographical centre of gravity of demand will continue to shift away from the Atlantic Basin towards the Pacific. This changing location, which will not rule out cyclic fluctuations, carries both opportunities and risks for the mining companies and first stage mineral processors. Before we explore the possible impact on corporate structures we need to examine some aspects of supply.

Some issues on the supply side

Even if due allowance is made for increased recycling, the amount of new mine capacity needed for virtually all minerals will be much greater than the prospective increases in demand. This is because of the inevitable depletion of the ore deposits on which existing mines are based. As a broad generalisation, it can be said that roughly half the capacity that was in place for non-ferrous metallic ores 25 years ago has by now been replaced. This replacement, in many instances, has been effected through the provision of additional equipment to cope with deeper and less accessible ore from the same deposits; but without such expenditures, the mines would have had to close down.

Cyclical growth

Investment in new capacity does not grow smoothly through time, but it moves lumpily. There is evidence of investment cycles of 10 to 15 years' duration which interact with much shorter cycles in demand. These investment cycles have also been exacerbated by a changing political climate. Many of today's major mining operations first started during the period of rapidly growing demand of the decades up to the 1973–74 climacteric. These mines are now distinctly middle-aged, and in some instances nearing the end of their productive lives, already extended by capital investment. Typically, the costs of any mining operation soar as it nears exhaustion. During the coming decades, the world will have to replace these worn-out facilities as well as meet any growth in demand.

Depletion of high-grade ores?

Simpler models of mineral development assume that the higher grade and more accessible ore deposits will be exploited first. This is partially true, but only for any given state of technical knowledge, and within the confines of prevailing political and social constraints. The long run history of copper seemingly demonstrates that the average grades extracted have gradually declined. Yet copper may not be typical, and the trend is far from consistent.

Many recent mineral discoveries, even in previously well-explored and 'politically safe' countries like Canada, are as accessible as, and have higher grades than, those that have been worked for many years. Their metallurgy is also similar. Even before taking account of changing political and environmental risks, which at present are pulling in different directions, there is no clear tendency for the underlying real costs of minerals to rise. That is also before allowing for the effects of technological change and of the mining companies' productivity improvements.

Costs, prices and derivatives

In recent years most mining companies have concentrated relentlessly on reducing their cash operating costs through every available means. Both the average and marginal costs of production of mineral products have fallen in real terms. A prolonged game of competitive leapfrog is being played in which relative costs are regarded as more important than their absolute levels. Some of the advantages of relatively low production costs are being offset, at least in the short to medium term, by the growing use of more complex means of price protection through the markets for derivatives, and particularly through complex options. These are many headed hydra, whose potential impact on competitiveness and on future balances between supply and demand may not be fully appreciated by technically dominated mining companies. Nor indeed have the industry's analysts properly explored their potential implications. Is a resort to derivatives ultimately a zero sum game that benefits the intermediaries who sell them, rather than the producers and consumers who use them? Stock market ratings of gold companies suggest otherwise—although stock markets are not noted for their long term vision. The various scandals of recent years have highlighted the risks involved, but have merely scotched the expanding use of price derivatives.

Technological innovations and economies of scale

Aside from changes in processes, such as those which have altered the structure of the gold and copper industries, the diffusion of technological innovations has helped reduce costs. One of the most pervasive trends, however, has been the steady exploitation of technical economies of scale. Especially in open-pit mining, higher throughput has led to ever lower costs. Such economies of scale have extended to transport, which has been most beneficial to the bulk minerals, such as iron ore. The technical advantages of scale have to be balanced against corresponding costs. The larger the mine, the more intrusive it becomes on the local environment and surrounding community; the greater its economic impact on the host country, the larger the capital commitment required, and the more it needs to operate at full capacity in order to remain profitable. Many of these other costs have not so far been internalised, but companies are facing mounting pressures to bear them more directly. To the extent that these pressures are effective, the unrelenting trend towards ever larger and more intrusive mines will be at least modified.

Not all minerals and metals are amenable to large scale open-pit methods of mining. Underground operations generally have less pronounced impacts on the local environment. Moreover, SX-EW[1] operations in the copper industry were initially applied on a relatively small scale. Are similar technological changes, such as the pressure-leaching of nickel laterites, likely to affect other minerals, and change the structure of their costs and the technically optimum scale of mines?

Further cost saving devices

One of the more obvious operations in which to seek cost savings is in the transport of ores and concentrates from mines to processing plants. Over half the material transported is normally waste of one form or another, usually including a fair amount of water. Will new processes for extracting both ferrous and non-ferrous metals move from the pilot plant stage over the next two decades into full commercial operation? Several are already well advanced, having already been under development for decades. All will carry technical and commercial risks, but these are very different from those presently borne, often unknowingly, by large open-pit operations.

Contracting out

Mining companies have followed a general trend towards contracting out operations of all types in order to reduce their overheads, and increase their flexibility

[1]Leaching techniques using solvent extraction (SX) and electrowinning (EW).

of response to changing conditions. The costs of hiring, employing, and more relevantly firing, workers fall on the contractors. Services of all types were usually the first put out to tender. More recently, and particularly in Australia and nearby countries, maintenance and mining have been contracted out. Contract mining companies now account for a significant and rising share of Australia's mineral production. This is a poorly documented area, on which there are few, if any, reliable statistics. Contract mining enables smaller companies to obtain many of the benefits of economies of scale without commensurate capital investment or labour forces.

In many cases, contracting is associated with 'fly in, fly out' operations which have only a small permanent staff on site, and consequently do not need the large social infrastructure that would be required by a large local labour force and their families. Such 'commuter' mining is also common in Canada. Will it become the norm for mines in remote or environmentally sensitive areas? It is clearly one means of reducing a mine's footprint on its immediate environment, and of lowering the eventual costs of closure and rehabilitation.

Even if contract mining is eschewed, companies are increasingly able to lease both mobile and fixed equipment, whether from the equipment suppliers, or from specialist leasing companies. This practice has been taken to its extremes by the airlines who seldom own their aircraft outright. Again the effect is to reduce the capital spending required up front.

Corporate structures

Multi-nationals or smaller companies?

Is the mining industry moving towards the emergence of the virtual mining company? Not according to prevailing viewpoints. There is a widespread belief that the industry will be increasingly dominated by a few successful multi-national companies. These MNCs will be surrounded by a penumbra of smaller companies, whether national or international, many on the verges of profitability. The recent relative growth of companies like Rio Tinto, BHP, and Gencor is adduced in evidence. In practice the global industry's overall concentration is not materially different from what it was 20 years ago, as the analyses of the Swedish Raw Materials Group have made clear. What has changed is the actors involved. The structure of ownership has been an evolving kaleidoscope, which is likely to continue changing, and in unexpected directions. Those who believe otherwise are often misled by developments in specific sectors. They also confuse the scale of mines with that of mining companies. Certainly a few large mines have contributed disproportionately to the industry's total profits, but that has always been the case, and the absolute size of profits is not the same as profitability. Moreover, the preceding discussion has raised doubts about whether the trend towards increasing scale of individual operations will necessarily persist.

During the 1950s, the global mining industry was dominated by a few predominantly North American companies. That dominance was expected to remain unchallenged, but many of the major companies of that era have since been taken over, or have completely disappeared. During the late 1960s and throughout the 1970s, the prevailing trend was towards rising state ownership, as foreign owned assets were nationalised. In the developed countries like Australia and Canada this resources nationalism was asserted through ever more onerous restrictions on the foreign ownership of mineral assets. There were fears that international mining companies would be squeezed out of business, and that there would be an absolute dearth of investment leading to a scarcity of resources. During the late 1970s and early 1980s, the remaining private sector mining companies were fast falling under the control of the major oil companies. That further added to the mining industry's fears for its future.

Subsequent developments show that seemingly irreversible trends are nothing of the sort. A more recent, and perhaps more parochial, example is provided by the zinc industry. Mergers and consolidations in the late 1980s strongly suggested, particularly to those involved, that the international industry was being concentrated in the hands of three or four large companies who could effectively control prices and markets. Recent history has proved otherwise. Many new developments have come from smaller, nimbler, or more creative companies.

Corporate size versus energy and adaptability

Whether or not the major international companies maintain, or even increase, their share of the industry's profits does not just hinge on developments outside their control. It also depends on their energy and adaptability. Neither sheer size nor growth is sufficient, but they have to be accompanied by continued profitability, which is by no means the same thing. There may be potential economies of scale at the corporate level in raising finance, and diffusing technology, but they are nowhere near as great as the technical economies achievable at individual operations. The latter can be accessed, as many companies are discovering, through joint ventures and other co-operative investments. Rising corporate size carries the dangers of bureaucratic rigidity and managerial inflexibility. As the past has repeatedly shown, this year's investment fashion can turn into tomorrow's outcast.

Geographic patterns of investment

Liberalisation and deregulation

Nowhere is fashion more evident than in the geographic location of the mining industry's exploration and investment. One of the clearest trends of recent years has been the liberalisation and deregulation of economic activity throughout the world, and the acceptance of foreign direct investment that has accompanied it. In the early 1980s foreign investment in mining was restricted mainly to Australasia and North America, and even then usually on sufferance. Today there are very few regions or countries that are off limits, at least in theory. One of the main thrusts has been to reduce the unacceptably high burdens of public sector deficits through privatising state-owned productive assets. That was the initial impetus even in the United Kingdom, with the superstructure of ideological commitment only being added later when the process was already well under way. Elsewhere it has been grafted onto completely different national traditions and inclinations.

Foreign direct investment: attitudes

A results-oriented approach to private ownership carries implicit dangers. It will remain acceptable only in so far as it delivers the expected benefits. Attitudes to foreign ownership move in cycles, often of an extended duration. Recent trends have clear parallels in the nineteenth century. The original Rio Tinto Company of the 1870s was formed by British investors, for example, to take over some moribund state-owned assets that had become an unacceptable drain on the Spanish exchequer. A few decades later, and in reaction to the company's success, widespread resentment had built up over foreign control of Spanish patrimony.

The recent growth of foreign direct investment in the minerals industry is most advanced in Latin America, but other regions are catching up. It was initially concentrated in gold, and in many countries it still is. The cost structures, capital requirements, weight to value ratios, and risk/reward profiles of gold production all contribute to its pioneering role. Investment in copper has followed, where the geologic climate is favourable. In that regard, it is a moot point whether foreign companies would have been as active had copper markets not been as tight over the past decade and prices as high relative to marginal costs. Strong demand has encouraged its own supply.

Political/legislative framework

To what extent are the political and legislative changes of recent years deep-seated or opportunistic? Are they just the latest fashions embraced by governments anxious to secure IMF/World Bank approval and attract foreign capital in order to solve immediate national problems? In many countries the foundations of the new policies appear shallow, and liable to shake when the immediate needs have passed. An apparently welcoming government, a suitable mining code, and an internationally competitive tax system are all necessary, but they are nowhere near sufficient to attract and retain stable long-term investment. Everything depends on the way in which laws and regulations are applied in practice, on the administrative and bureaucratic systems in place, and on the behaviour of the companies themselves. Many countries that have recently altered their mining and tax regimes are still unlikely to attract much investment into their mineral industries because the other prerequisites are lacking.

Political risk

Political risk will remain a major impediment to widespread direct foreign investment, even in countries with apparently rich natural resources. In many instances there are deep-seated structural, ethnic, or social problems which can only be overcome, if at all, by strongly committed governments which have the support of a considerable percentage of the population. Even so, deep-running undercurrents in society may still well up later to swamp foreign investment. Potential investors in many countries have to be fully aware of such possibilities which are seldom, if ever, reflected in conventional assessments of political risk. Governments, even when freely elected, are not always closely in tune with the social and cultural values of a large proportion of the potential electorates. Look, for example, at the strong underlying Australian desire for the country not to be treated just as a quarry, but as a country which 'adds value'. This latent desire periodically surfaces in official policies. The sentiments are by no means confined to Australians, but are common wherever mineral resources are exploited. Unless mining companies respond positively to those aspirations they will again run into the type of opposition that boiled over into the nationalisations of the post-war decades.

Downstream operations

Over the very long term there has been an interesting tendency for mining to be followed some 15 to 20 years later by first stage processing. Such a lag has been evident across different minerals and different countries. The trend may have been distorted in recent decades by the persistence of Europe's smelting industries, and by the voracious appetite of Japanese industry for raw materials. The satisfaction of that appetite by imports of ores and concentrates rather than metal was helped by protective tariffs, and by the relatively low operating costs of Japanese plants in a period of excess supply.

Although other Asian economies have started to follow the Japanese path, those conditions may be changing. Not only has excess processing capacity been largely absorbed over the past decade, but the industrialising countries no longer see the need to go through the same capital-intensive pattern of industrialisation that begins with heavy industry. Harking back to our earlier discussion of corporate structure, direct investment in overseas mines and processing plants by resource-deficient industrialising countries is rising rapidly. The People's Republic of China and the Republic of Korea are prominent examples. In some instances they are the sole investors, but more commonly they are taking local partners. These are seldom, if ever, the major international companies, but smaller and often local firms. This partial, and growing, counterbalance to the power of the major companies is an interesting trend to watch.

Long-term relations with host countries

Those mining companies that are not willing, or able, to make more than token gestures to local aspirations for further processing are liable to lose out to their more compliant rivals. Downstream investments that may not make much economic sense in the shorter term may be part of the price of a continued presence in a country. Another may be a degree of concerned involvement in the national life of the host country. At some stage it may be necessary to attract local shareholdings, and not just by a small elite which is close to the government of the day. No matter how the international climate has changed, the concept of the obsolescing bargain remains as relevant as ever. There is nothing more captive than a mining company with a sunk investment. Foreign investors will always be readily available scapegoats for national ills, irrespective of their legal position and of the possibility of international sanctions. That is especially the case where they are closely identified with an unpopular or unrepresentative government, or with a specific clique within that government. Personalities invariably change, and this year's top dogs can easily become tomorrow's outcasts.

The enclave syndrome

These issues are most acute in the remaining kleptocracies that are vying with each other for foreign investment. The pendulum has recently swung strongly in favour of foreign investors in most countries, but it is important to recognise that there is a pendulum. Where it settles in any country depends partly on global trends and fashions, but mainly on internal social and economic conditions. It will also be affected by the behaviour of the mining companies themselves.

Governments and their constituents may have grossly unrealistic expectations of foreign investment, both as regards the nature of any benefits and the likely pay back periods. Unless those expectations are deflated before any funds are committed even on exploration, trouble will be stored up for the future. Mining seldom creates much employment directly, but it is potentially a source of tax revenues. Part of the economic rent that accrues from mineral developments needs to be invested in assets that can eventually earn sufficient revenue to replace depleted ore deposits. That investment can be made either by the mining companies or, more probably, by the host governments. If it is made by the latter it has to come out of tax revenues. One important, if unpopular and even heretical, question is whether competitive bidding for a limited amount of investible funds has driven tax rates below the levels required for the long-run sustainability of mining in individual countries. The answer very much depends on how the various social and environmental issues are tackled by all concerned, as well as on the strictly economic questions of depletion and asset replacement.

Environmental and social issues

The appropriate division of any economic rent is only one aspect of the mining industry's long-run sustainability. As the frontiers of mineral exploration are pushed back, mining companies are increasingly moving to remote areas which have previously had little contact with modern civilisation. The clash of cultures and values can be dramatic, especially where large scale projects are involved. Such clashes are not specific to the mining industry, whatever many of those involved might think, but affect any large-scale modern development in a backward area. Mining is, however, more visually intrusive, and has a more long-term, if strictly localised, impact on land-use than most other types of investment. It also offers a prospect of riches beyond the dreams of avarice that is not afforded by other intrusive developments like hydropower schemes. There is potential economic rent to be captured by the local communities. Large scale open-pit mines and their associated infrastructure have the most obvious, if not the most insidious or damaging, impacts. It is a grave mistake for the mining industry to believe that local communities are solely interested in squeezing the maximum possible income from new mineral developments. That can certainly be a strong motive, but concern over the local environment and with the preservation of traditional cultures and patterns of living are often much more important. Failure to understand, and to respond sympathetically to, these concerns can disrupt mining projects and inhibit their effectiveness, regardless of any agreements reached with national governments. In that regard, the fate of Bougainville Copper in Papua New Guinea provides a cautionary tale.

New mine developments can be more readily absorbed into places which have many mineral deposits and traditions of mining, than into rural or wilderness areas with no such traditions. In order to gain even a modicum of acceptance, all mining companies, large or small, will have to adopt the highest possible environmental and social standards, no matter where they are investing. There is an increasing tendency to internalise the costs both of environmental damage and of the ultimate rehabilitation of mine sites, through a mixture of fiscal charges and regulation. The more that happens, the more mining companies will directly face a mounting conflict between the costs of environmental protection and technical economies of scale. The appropriate response in this instance is unlikely to be a move to even larger scale. Other possible impediments to increasing scale, and countervailing technical or economic changes that are now under way, have already been mentioned.

One common response to the claims of local communities is that mining companies should take direct responsibility for their health, education, housing, and economic advancement. Such an attitude has been taken to extremes in the former Soviet Union, where mineral enterprises have been indistinguishable from local government. The disadvantages in that context are so self-evident that it is a wonder that proponents of this approach do not see the lessons for elsewhere. The declining competitiveness of the Zambian copper industry over many years offers another cautionary tale. The prime function of mining companies is to develop mineral deposits competently and efficiently, and to maximise profitability over the long term within the constraints set by society.

There are many dangers in a paternalistic approach. The mining companies gradually usurp the proper functions of government, whether national or local, and create a culture of dependency. That may be narrowly acceptable when sizeable economic rents are available from a project, but it is not when profitability is squeezed by adverse market conditions. The companies then carry a superstructure of social costs and obligations which cannot be quickly shed. In consequence they are unable to invest adequately, and they then lag behind in the march towards greater competitiveness. That is what happened to the Zambian copper industry. Furthermore, undue expectations are aroused about the proper functions of mining companies which inhibit investment in marginal operations that can never bear such costs. In time the companies have to shed many of their assumed social costs if they are to survive in a competitive industry and thus maintain employment and incomes. In that regard the experiences of Rossing Uranium in the early 1990s are salutary. Market conditions forced a retreat from its previous all-embracing paternalism. Its policy was changed before it had become so entrenched as to create an enclave economy that was increasingly divorced from the remainder of the economy. The fate of the enclave cultures of Chile's American-owned copper mines in the 1950s and 1960s illustrates all too clearly some of the potential dangers of corporate paternalism. The mining companies are themselves no substitute for efficient and effective government at all levels. The wealth they create can provide the means for society, working through governments, to meet social needs, but they run grave risks if they usurp the functions of government, however well-intentioned they might be.

A more effective approach is to structure projects in such a fashion as to minimise their potentially disruptive impact on a local environment and culture, and to maximise the benefits for the local community. Here, Rossing Uranium provides a valuable role model. From the outset, a percentage of its net profits was contributed to an independent foundation whose objective was the development of the local community, broadly defined.

Concluding remarks

The mining industry currently finds itself increasingly squeezed by environmental concerns, covering the whole range of operations from grass-roots exploration to final end-use of mineral products. This development may be inextricably linked to rising standards of living, and the associated trend towards paying more attention to the quality of life, as basic material needs gradually become satisfied. Rich societies can afford to minimise the disruptive impact of development on the environment, even to the extent of foregoing the financial benefits of that development. The same standards cannot, however, be applied in countries still blighted by poverty and disease; such impositions by well-meaning elements within rich countries are resulting in negative side-effects, not only for the mining industry.

The industry has to adopt the highest possible health, safety, and environmental standards that are consistent with its long-run viability, wherever it operates. It constantly has to distinguish between extreme views and reasoned legitimate concerns. It can, and should, do no more.

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